DevOps Infrastructure – Unit 2 Notes (Part 5)

Copy-on-Write Mechanism (Storage Perspective)

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1. Copy-on-Write (CoW) – Concept

Definition

Copy-on-Write is a mechanism where:

- Original data is not modified
- Changes are written to a new layer

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2. How It Works in Docker

- Image layers are read-only
- When a container starts:
    - A writable layer is added on top

Behavior

- Read → from image layers
- Write → stored in container layer

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3. Example

If multiple containers use the same image:

- All share base layers
- Each has its own writable layer

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4. Advantages

- Efficient memory usage
- Faster container startup
- Reduced duplication of data

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5. Limitation

- If container is deleted, writable layer is lost
- Requires volumes for persistence

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6. Summary

- Copy-on-Write ensures efficient storage
- Containers do not modify base images
- Writable layer stores runtime changes
- Persistence requires external storage